

Enrolment No: _____

Name of Student: _____

Department/ School: _____

END TERM EXAMINATION ODD SEMESTER 2022-23**COURSE CODE: CSET203****MAX. DURATION: 2HRS****COURSE TITLE: Microprocessor and Computer Architecture****TOTAL MARKS: 35****COURSE CREDIT: 4****GENERAL INSTRUCTIONS: -**

1. Do not write anything on the question paper except **name, enrolment number** and **department/school**.
2. Carrying mobile phone, smart watch and any other non-permissible materials in the examination hall is an act of **UFM**.
3. Write **batch number** on the answer sheet.

SECTION A**Q1. Attempt all questions.****(1 x 5 = 5 Marks)**

- a) Imagine a smartphone consisting of Arm Cortex 7 processor. With this statement determine the number of bits utilized by the CPU and its ALU.
- b) Determine the pipeline's efficiency (in %) if it is a 5-stage pipeline with a delay of each 20 ns, 30 ns, 40 ns, 50 ns, and 60 ns.
- c) A hard disk with a transfer rate of 10 MBps is constantly transferring data to memory using DMA. The processor runs at 600 MHz and takes 300 and 900 clock cycles to initiate and complete DMA transfer, respectively. If the size of transfer is 20KB. What is the percentage of processor time consumed for the transfer operation?
- d) Consider an 8-way set associative mapped cache. The size of cache memory is 512 KB and there are 10 bits in the tag. Find the size of main memory.
- e) Define the following terms: (i) Stall (ii) Average Cache Memory Access Time

SECTION B**Q2. Attempt all questions.****(3 x 5 = 15 Marks)**

- a) What kind of controller will be used if you want direct access to memory and I/O devices? Explain the architecture with an appropriate diagram.
- b) List the three differences between each of the following.:
 - (i) RISC and CISC
 - (ii) CPU and GPU
 - (iii) Hard disk drive (HDD) and solid-state drive (SSD)
- c)
 - (i) Consider a fully associative mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find the number of bits in tag and tag directory size
 - (ii) Take into consideration a 2KB set associative cache with 64-byte cache blocks. Assume that a 32-bit address is used to reach the cache, which may be accessed using byte addresses. The associativity of the cache should be determined if the tag field's width is 22 bits.
- d) Assume the following instructions.
I1: R1 = 100
I2: R1 = R2 + R4

$$I3: R2 = R4 + 25$$

$$I4: R4 = R1 + R3$$

$$I5: R1 = R1 + 30$$

Calculate sum of (WAR, RAW and WAW) dependencies the above instructions.

- e) A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decodes (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX-stage. The sequence of instructions corresponding to the statement $X = (S - R * (P + Q)) / T$ is given below. The values of variables P, Q, R, S and T are available in the registers R0, R1, R2, R3 and R4 respectively, before the execution of the instruction sequence.

ADD R5, R0, R1;	$R5 \rightarrow R0 + R1$
MUL R6, R2, R5;	$R6 \rightarrow R2 + R5$
SUB R5, R3, R6;	$R5 \rightarrow R3 - R6$
DIV R6, R5, R4;	$R6 \rightarrow R5 / R4$
STORE R6, X;	$X \leftarrow R6$

The IF, ID and WB stages take 1 clock cycle each. The EX-stage takes 1 clock cycle each for the ADD, SUB and STORE operations, and 3 clock cycles each for MUL and DIV operations. Determine the number of clock cycles required to complete the sequence of instructions.

SECTION C

(5 x 3 = 15 Marks)

Q3. Attempt all questions.

- a) Imagine a pipeline system that has four stages: instruction fetch (IF), instruction decodes (ID), execution (EX), and write back (WB). The following table illustrates the number of clock cycles needed for each instruction to finish each stage. How many clock cycles and stall are required to complete the instructions?

Instructions	Instruction Fetch (IF)	Instruction Decode (ID)	Execution (EX)	Write Back (WB)
I ₀	1	1	2	1
I ₁	2	2	3	1
I ₂	2	2	2	2
I ₃	2	1	1	1
I ₄	3	2	1	2

- b) ARM 7 processor are used in modern mobile processor, draw and explain the design of the architecture of ARM7 processor in terms of Data path design.
- c) Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI, and WO are 4, 7, 9, 10, and 11 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 2ns. A program consisting of 14 instructions I₁, I₂, I₃, ..., I₁₄ is executed in this pipelined processor. Instruction I₄ is the only branch instruction, and its branch target is I₁₀. If the branch is taken during the execution of this program, then estimate the time (in ns) needed to complete the program. Also, include the stalls clock cycles to estimate the time.

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SECTION B**Q2. Attempt all questions.****(3 x 5 = 15 Marks)**

- a) What kind of controller will be used if you want direct access to memory and I/O devices? Explain the architecture with an appropriate diagram.
- b) List the three differences between each of the following.:
 - (i) RISC and CISC
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STORE R6, X;	$X \leftarrow R6$

The IF, ID and WB stages take 1 clock cycle each. The EX-stage takes 1 clock cycle each for the ADD, SUB and STORE operations, and 3 clock cycles each for MUL and DIV operations. Determine the number of clock cycles required to complete the sequence of instructions.

SECTION C

Q3. Attempt all questions.

(5 x 3 = 15 Marks)

- a) Imagine a pipeline system that has four stages: instruction fetch (IF), instruction decodes (ID), execution (EX), and write back (WB). The following table illustrates the number of clock cycles needed for each instruction to finish each stage. How many clock cycles and stall are required to complete the instructions?

Instructions	Instruction Fetch (IF)	Instruction Decode (ID)	Execution (EX)	Write Back (WB)
I_0	1	1	2	1
I_1	2	2	3	1
I_2	2	2	2	2
I_3	2	1	1	1
I_4	3	2	1	2

- b) ARM 7 processor are used in modern mobile processor, draw and explain the design of the architecture of ARM7 processor in terms of Data path design.
- c) Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI, and WO are 4, 7, 9, 10, and 11 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 2ns. A program consisting of 14 instructions $I_1, I_2, I_3, \dots, I_{14}$ is executed in this pipelined processor. Instruction I_4 is the only branch instruction, and its branch target is I_{10} . If the branch is taken during the execution of this program, then estimate the time (in ns) needed to complete the program. Also, include the stalls clock cycles to estimate the time.